Reg. No. | | | | | | | | | | | [14EC61]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

DIGITAL COMMUNICATION

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing at least TWO questions from each part.
2. Q-function table / ERC function table may be used if necessary
3. Missing data may be suitably assumed.

PART - A

1. a. With a neat diagram, explain a digital communication system and indicate the function of each block. **10**

b. Specify the Nyquist rate and Nyquist Interval for each of the following signals **04**

i. $x(t) = {}^{1}/_{2\pi}\text{COS}(4000\pi t)\ \text{COS}\ (1000\pi t)$

ii. $x(t) = \text{SINC}\ 500\pi t\ /\pi t$

2. a. With a suitable example, distinguish between the uniform and non-uniform quantization in a PCM technique. **05**

b. Write notes on Companding. **04**

c. The information in an analog waveform with maximum frequency fm =4KHz is to be transmitted over an M-Level PCM system where the number of pulse levels is M=16. The quantization distortion is specified not to exceed 1% of peak to peak analog signal. **05**

i. What is the Max number of bits /sample that should be used in this PCM system?

ii. What is the Minimum sampling rate? What is the resulting bit transmission rate?

3. a. What is the slope over load distortion and granular noise in Delta Modulation? How it can be removed in ADM? **10**

b. Sketch the given digital data 1 0 1 0 1 0 11 0 0 in **04**

i. Unipolar RZ

ii. Bipolar NRZ

iii. Manchester coding

iv. Bipolar RZ line coding formats

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4. a. Discuss in detail about Inter Symbol Interference (ISI) and the measures to reduce the same. **10**

b. The binary data **0 0 1 1 0 1 0 0 1** are applied to the input of a modified duobinary system. **04**

i. Construct the modified duobinary coder output and corresponding receiver output, without precoder.

ii. Suppose that due to error during transmission, the level at the receiver input produced by the third digit is reduced to zero. Construct the new receiver output.

PART - B

5. a. With the help of a diagram and waveform, how a message is transmitted in QPSK? **10**

b. Sketch the in-phase and Quadrature components of a QPSK signal for the binary sequence 110010111. Assume carrier frequency fc to be equal to 1/Tb choose any convenient basis functions. **04**

6. a. With a neat diagram and a suitable example, explain the working of a DPSK transmitter and DPSK receiver. **10**

b. Determine the minimum bandwidth and baud for BPSK modulator with a carrier frequency of 40MHz and an input bit rate of 500kbps. Sketch the output waveform. **04**

7. a. Derive an expression for a Probability of error in a BPSK coherent receiver. **10**

b. Binary data is transmitted at a rate of 10^6 bits/sec over microwave link having a bandwidth of 3MHz. Assume that the noise power spectral density at the receiver input is $\eta/2 = 10^{-10}$ watt / Hz. Find the average carrier power required at the receiver input for coherent PSK and DPSK signaling schemes to maintain $P_e \leq 10^{-4}$. **04**

8. a. A 3-stage shift register with linear feedback generates the sequence. 01011100101110. Determine the period of the given sequence. Verify the three properties of the PN sequence for the given sequence. **06**

b. What is spread spectrum modulation and bring out the significance of PN sequence in Direct Sequence Spread spectrum technique. **08**